

Academic Writing

A short guide to the process, the structure, and the introduction

Irina Gaynanova

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1 Writing is a skill

Academic work involves a great deal of writing: papers, grant proposals, application materials, software documentation, dissertation chapters. All of it can be learned and improved like any other skill — calculus, programming, gardening. It is also hard, especially if you care about quality, and it is mostly a matter of **discipline** rather than talent.

Four practices do most of the work:

- **Plan regular time blocks**, even short ones. Thirty minutes on a schedule beats waiting for a long stretch.
- **Separate generating text from editing**. Generating is about 20% of writing; editing is the other 80%. Doing both at once stalls you on the first sentence.
- **Embrace the bad first draft**. Its only job is to exist. You cannot edit a blank page.
- **Plan for repeated feedback**, and expect to edit, edit, edit as a result.

What not to do:

Anti-pattern	Why it fails
Write one sentence and rewrite it five times	Editing before there is anything to edit
Wait for a free three-hour block	It rarely comes; short blocks accumulate
Wait for inspiration	Writing is discipline, not mood
Give up because you are “not good at it”	It is a skill, and skills respond to practice
Polish grammar before asking for feedback	Wasted effort on text that may be cut

Excellence has two halves. The **process**: dedicated time and focus, generation separated from editing, planned feedback and multiple rounds of edits, version control (Overleaf for LaTeX, Google Docs for Word). The **product**: it follows the expected structure with all required components; its style is tailored to the audience; it aims to educate rather than confuse, with clear word choices and simple sentences and no hidden steps; and it is concise and concrete — no watery sentences, no filler, claims substantiated with references.

2 Structure

Two structures cover most cases. Decide the target journal first, then find two or three example papers from it to guide structure, style, and language.

Statistical methodology journals	Life sciences / domain journals
Introduction	Introduction
Methods (may be several sections)	Methods
Simulation studies	Results
Application to X study	Discussion (often long)
Discussion / Conclusions	

Machine learning conference papers usually merge simulation and data analysis into a single Numerical Comparison section, since benchmark datasets are standard there rather than simulated data.

What each section does:

Methods describes what you are going to do and how — *not* the results. In a domain paper: study design (data source, collection, what was collected), data processing, statistical methods. In a methods paper: study design and processing if relevant, then the proposed model and estimation procedure (the core), then theoretical investigation if applicable.

Results presents findings without interpreting them. In a domain paper: Table 1 with sample characteristics, exploratory analysis as applicable, then results from fitted models. In a methods paper this is two sections — Simulation studies, which always covers data generation, methods for comparison, metrics for comparison, and results; and Application to X study, which covers study design and processing (if not given earlier), methods for comparison including the proposed one, and results.

Discussion gives a two-to-three sentence summary, then strengths and limitations, then opportunities for future research. Domain papers discuss at length, especially limitations and scientific implications. Statistics papers are shorter — often a single paragraph, focused on the method and on future improvements.

Abstract, typically 250 words or fewer, mirrors the paper: introduction, methods, results, conclusion. It must be self-contained, so that a reader gets the idea of the paper without reading it.

3 The introduction

The introduction moves from broad to narrow and must hit four things: the context of the problem and why anyone should care; existing approaches and why they fall short (or what gaps remain in what is known); the new thing you will do about it; and an explicit statement of the work's contributions.

Paragraph by paragraph, a typical version:

Paragraph	Job
1	Context for the problem. For a data-centric paper, connect to the data immediately. Establish that this is a setting worth looking at.
2–3	Literature review: what has been done. The punchline is the gap or the limitation.
4	Motivate your approach from that gap. The big sell — what is cool about it, how it overcomes the limitation.

Paragraph	Job
Last	Outline of the paper: “The rest of the paper is organized as follows. In Section 2, we...” — one line per section.

Journals differ on that last paragraph. Most expect it; a few explicitly ask that it be left out. Check before you submit.

Real introductions vary around this template, and reading several of your target journal’s is the fastest way to calibrate. A theoretical paper may go straight to a mathematical statement of the problem and close with a point-by-point contributions list. A short computational paper may spend two paragraphs on why the problem matters, one on the limitation of the state of the art, and one on how it is solved. An applied paper may open with the data type and concrete data questions before reviewing methods at all.

4 Why papers get rejected

Most rejections trace to one of six reader reactions. Each has a fix, and each fix belongs mostly in the introduction.

The reader...	What addresses it
doesn’t think the work is novel	Acknowledge existing work properly, and say clearly how yours differs and what is new
doesn’t see the problem being addressed	Articulate the specific question or problem the work solves
sees the problem but doesn’t think it matters	Connect to the scientific domain through consequences — if this is not addressed, then what?
doesn’t understand what you did, or why it solves the problem	State contributions explicitly, contrast with prior work, present results so they visibly solve the problem
doesn’t think it fits the audience	Ask who cares most, whether it matches the journal’s aims and scope, and whether it is written for that reader
finds the work sloppy	Feedback, multiple edits, proofreading

Feedback checklist

Give this to a reader — and use it on yourself. After reading only the abstract and introduction, they should be able to check every box:

- ☐ I clearly understand the problem the work addresses, or the knowledge gap it fills
- ☐ I clearly see why addressing this problem is important
- ☐ I understand the approach the authors took
- ☐ I am convinced the approach differs from what has been tried, in meaningful ways
- ☐ I am convinced the approach succeeds in addressing the problem
- ☐ I found the writing easy to follow, and I learned something from reading it

Practical notes

Find model papers. Identify the target venue first, then look through recent issues for two or three well-written accepted papers resembling yours. Use them to judge how introductions are written, what goes in the discussion, and how much detail belongs in the main body versus the appendix. Ask a senior author whether the ones you picked are good guides.

Write in active present voice. *We consider...*, *we apply our method to...*, *we compare...*; likewise *Figure 1 shows*, *Table 1 shows*. The exception is a domain journal whose field prefers the passive.

Use the LaTeX template. The lab's LaTeX template document covers labels and cross-referencing, mathematical notation conventions, bibliography management with BibTeX, and tables, figures, and algorithms. Two rules from it worth repeating here: generate LaTeX tables from R with `xtable` rather than copying numbers by hand, and save figures with the `pdf()` command rather than the RStudio export button.

Get help. The [SPH Writing Lab](#) and the [Sweetland Center for Writing](#) both offer support.

Further reading

Strunk W, White EB. *The Elements of Style* — grammar and style.

Silvia PJ. *How to Write a Lot: A Practical Guide to Productive Academic Writing*. APA LifeTools — motivation and persistence.

Goodson P. *Becoming an Academic Writer: 50 Exercises for Paced, Productive, and Powerful Writing*. SAGE — exercises, and advice on specific sections.

Lamott A (1994). *Bird by Bird: Some Instructions on Writing and Life*. Anchor Books — the chapter on first drafts is the classic statement of why bad first drafts are the only route to good final ones. See also the Publication Coach's "[The joy of the crappy first draft](#)".

Ecarnot F, Seronde M-F, Chopard R, Schiele F, Meneveau N (2015). Writing a scientific article: a step-by-step guide for beginners. *European Geriatric Medicine*, 6(6), 573–579 — a practical walkthrough aimed at first-time authors.

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